

AP-E9 Scaling Trichotomy and Regulator-Stability Audit: Equicoercivity, Gamma Limits, Alternative Triangulations, and Translated Starts

Route F research note

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Abstract

This note resolves the first mathematical layer left open by AP-E8 and gives a negative result for the stronger layer. On a fixed physical box, the original coordinate-edge Dirichlet term already makes the full discrete action equicoercive in strong $L^2 \times L^2$, independently of the barrier scaling. This compactness is too weak to close a degree sector. Writing $d_a = 1 - \epsilon_a$, $w_a = \gamma_a a$, and $R_a = \gamma_a a^2 / d_a^2$, the logarithmic barrier has an exact Lambert- W edge-margin bound and the global estimate

$$b_{\epsilon_a}(n_x \cdot n_y) \geq \frac{|n_x - n_y|^4}{8d_a^2}.$$

For fixed ϵ and $\gamma(a) = ca^{-p}$, uniform distance from the edge floor requires $p \geq 1$, whereas the barrier disappears on smooth fields only for $p < 2$. Hence $1 \leq p < 2$, with the minimal choice $p = 1$, is the unique edge-interior and smooth-vanishing power window. Nevertheless $R_a \rightarrow 0$ supplies no C^0 compactness and the explicitly embedded barrier-only Gamma limit forgets trace and degree. At the critical value $p = 2$, the barrier does give $W^{1,4}$ compactness and closes degree. Its raw six-tetrahedron and checkerboard five-tetrahedron Cauchy–Born quartics are not scalar multiples. The actual finite- R Gamma density, however, requires a periodic homogenized cell problem and is not identified with either raw quartic in this note.

This is a no-free-lunch theorem for the AP-E8 barrier: no scaling makes that barrier both continuum-vanishing and, by itself, degree-sector compact. A full Gamma theorem for the same-action Skyrme model is therefore reduced to a precise remaining question about compactness of the discrete Skyrme minors and density of smooth fixed-degree recovery sequences. The accompanying card tests the recommended $p = 1$ trajectory, the critical $p = 2$ control, two checkerboard five-tetrahedron meshes, translated starts, and enlarged (a, L) ranges. The physical Hessian and determinant variation remain prohibited unless those tests and the missing minor-compactness theorem produce one regulator-stable continuum background.

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1 Claim ledger and notation

Let $\Omega \subset \mathbb{R}^3$ be a fixed cube and let $a \downarrow 0$ run over shape-regular cubic grids. A conforming tetrahedral complex $\mathcal{T}_a$ is placed on every grid, and $\mathcal{E}_a$ denotes the unique unordered vertex pairs that occur together in at least one tetrahedron. The nodal variables are

$$n_x \in \mathbb{S}^3, \quad s_x \in \mathbb{R},$$

with exact vacuum boundary data $n = e_0 = (1, 0, 0, 0)$ and $s = 1$. The AP-E8 functional is

$$\mathcal{F}_a[n, s] = E_a[n, s] + \mathcal{B}_a[n], \quad \mathcal{B}_a[n] = w_a \sum_{\{x, y\} \in \mathcal{E}_a} b_{\epsilon_a}(n_x \cdot n_y), \quad (1.1)$$

where

$$b_{\epsilon}(q) = \begin{cases} -\log \frac{q - \epsilon}{1 - \epsilon} + \frac{q - 1}{1 - \epsilon}, & q > \epsilon, \\ +\infty, & q \leq \epsilon. \end{cases} \quad (1.2)$$

The base action E_a is the AP-E7 coordinate-difference (n+s) action, including its nonnegative Skyrme-minor, breathing-mass, and pion-mass terms. We separate four statements that must not be conflated:

- (i) ordinary L^2 equicoercivity of all bounded sublevels;
- (ii) uniform distance from the barrier floor;
- (iii) compactness strong enough to preserve normalized-affine degree;
- (iv) Gamma convergence of the complete base-plus-barrier action.

This note proves (i), gives a sharp answer to (ii), proves the barrier compactness regimes controlling (iii), and gives a conditional reduction rather than an unsupported identification for (iv).

2 Exact barrier calculus

Put

$$d_a := 1 - \epsilon_a, \quad w_a := \gamma_a a, \quad R_a := \frac{\gamma_a a^2}{d_a^2} = \frac{w_a a}{d_a^2}. \quad (2.1)$$

For one admissible edge define

$$x_e := \frac{q_e - \epsilon_a}{d_a} \in (0, 1], \quad z_e := 1 - x_e = \frac{1 - q_e}{d_a} = \frac{|n_x - n_y|^2}{2d_a}. \quad (2.2)$$

Then the dependence on ϵ_a is isolated:

$$b_{\epsilon_a}(q_e) = \phi(x_e), \quad \phi(x) := -\log x - 1 + x. \quad (2.3)$$

Lemma 2.1 (Sharp elementary bounds). *For $0 \leq z < 1$,*

$$\phi(1-z) = \sum_{k=2}^{\infty} \frac{z^k}{k}, \quad \frac{z^2}{2} \leq \phi(1-z) \leq \frac{z^2}{2(1-z)}. \quad (2.4)$$

Consequently every admissible edge obeys the global lower bound

$$\boxed{b_{\epsilon_a}(n_x \cdot n_y) \geq \frac{|n_x - n_y|^4}{8d_a^2}}. \quad (2.5)$$

Proof. The Taylor series of $-\log(1-z) - z$ gives the equality and lower bound. For the upper bound, $1/k \leq 1/2$ for $k \geq 2$, hence

$$\sum_{k=2}^{\infty} \frac{z^k}{k} \leq \frac{1}{2} \sum_{k=2}^{\infty} z^k = \frac{z^2}{2(1-z)}.$$

Substituting $z = |n_x - n_y|^2/(2d_a)$ proves Eq. (2.5). $\square$

3 Sharp bounded-sublevel edge margin

Let W_0 be the principal real branch of Lambert W and define

$$\chi(r) := -W_0(-e^{-1-r}), \quad r \geq 0. \quad (3.1)$$

Because ϕ decreases from $+\infty$ to zero on $(0, 1]$, $\phi(\chi(r)) = r$.

Theorem 3.1 (Exact positive-sublevel margin). *Fix $C > 0$. If $\mathcal{B}_a[n] \leq C$, then every barrier edge satisfies*

$$x_e \geq \chi(C/w_a), \quad q_e - \epsilon_a \geq d_a \chi(C/w_a) \geq d_a e^{-1-C/w_a}. \quad (3.2)$$

For every fixed $C > 0$, uniform relative interiority $x_e \geq \delta_C > 0$ on the C -sublevels holds if and only if

$$\inf_a w_a > 0. \quad (3.3)$$

Uniform absolute interiority $q_e - \epsilon_a \geq \delta_C > 0$ additionally requires and is implied by $\inf_a d_a > 0$.

Proof. Nonnegativity of every summand gives $w_a \phi(x_e) \leq C$, and inversion gives the first inequality. Since $\phi(x) = -\log x - 1 + x \geq -\log x - 1$, the same sublevel inequality gives $x_e \geq e^{-1-C/w_a}$. If $w_a \geq w_* > 0$, the right-hand side is a positive number depending only on C, w_* . Multiplication by $d_a \geq d_* > 0$ proves the absolute statement.

For necessity, pass to any subsequence on which $w_a \rightarrow 0$. Choose one interior vertex in the exact vacuum collar and leave all other collar vertices equal to e_0 . Rotate that vertex on a great circle so that, for a fixed $\alpha > 0$ chosen smaller than C divided by twice the maximum incident-edge count,

$$n_z \cdot e_0 = \epsilon_a + d_a e^{-\alpha/w_a}.$$

The number of incident co-tetrahedral edges is bounded independently of a , and each contributes

$$w_a \phi(e^{-\alpha/w_a}) = \alpha - w_a + o(w_a).$$

The coordinate-edge Dirichlet cost is $O(a)$, the local potential cost is $O(a^3)$, and every affected forward Skyrme minor has rank at most one and therefore vanishes. Thus the field lies in the prescribed C -sublevel for all sufficiently small a while the relative edge margin tends to zero. For absolute necessity, if $d_a \rightarrow 0$, the vacuum itself has zero barrier and $q_e - \epsilon_a = d_a \rightarrow 0$. $\square$

Remark 3.2. The restriction $C > 0$ is essential to the equivalence: on the zero sublevel every connected-edge value has $x_e = 1$, independently of w_a .

Remark 3.3 (Margin is not the no-zero bound). If $\inf_a \epsilon_a > -1/3$, every finite-energy tetrahedron still obeys

$$\left\| \sum_{i=0}^3 \lambda_i n_i \right\|^2 \geq \epsilon_a + (1 - \epsilon_a) \sum_i \lambda_i^2 \geq \frac{1 + 3\epsilon_a}{4} > 0. \quad (3.4)$$

Hence $w_a \rightarrow 0$ destroys uniform distance from the artificial barrier floor, not the definition of degree on each individual grid.

4 Ordinary equicoercivity is already present

Let I_a denote the affine interpolant on any of the finitely many shape-regular triangulation families used below.

Theorem 4.1 (Fixed-box L^2 equicoercivity). *Assume, as in the declared AP-E7 action, that the coordinate-edge Dirichlet part is bounded below by*

$$c_D a \sum_{xy} (|n_x - n_y|^2 + |s_x - s_y|^2) \quad (4.1)$$

with a mesh-independent $c_D > 0$. On a fixed bounded cube with the declared boundary data, the family $\{\mathcal{F}_a\}$ is equicoercive in strong $L^2(\Omega; \mathbb{R}^4) \times L^2(\Omega)$ for every positive choice of γ_a and every $\epsilon_a \in (-1/3, 1)$.

Proof. Shape regularity gives a mesh-independent constant C_T such that

$$\|\nabla I_a n_a\|_{L^2}^2 + \|\nabla I_a s_a\|_{L^2}^2 \leq C_T a \sum_{\text{coordinate edges } xy} (|n_x - n_y|^2 + |s_x - s_y|^2) \leq \frac{C_T}{c_D} \mathcal{F}_a. \quad (4.2)$$

The first L^2 norm is bounded because the nodal values have unit norm. For $s_a - 1$, either Poincaré's inequality and the exact boundary value or the positive breathing-mass term supplies the L^2 bound. Rellich compactness gives strong $L^2 \times L^2$ precompactness. Finally,

$$\int_{\Omega} \text{dist}(I_a n_a, \mathbb{S}^3)^2 dx \leq C a^2 \|\nabla I_a n_a\|_{L^2}^2 \rightarrow 0,$$

so the limiting field is sphere-valued almost everywhere. The barrier is nonnegative and therefore cannot weaken these estimates. $\square$

Remark 4.2. Strong L^2 convergence does not preserve the degree of a three-dimensional sphere-valued map. Calling Eq. (4.2) “topological equicoercivity” would therefore be incorrect.

5 Smooth expansion and the power-law window

Suppose $n \in C^3(\overline{\Omega}, \mathbb{S}^3)$, and let $S_a n$ denote nodal sampling. For an edge $y = x + av$,

$$1 - n(x) \cdot n(x + av) = \frac{a^2}{2} |D_v n(x)|^2 + O(a^3).$$

If $a^2/d_a \rightarrow 0$, Eqs. (2.4)–(2.2) and a Riemann-sum estimate give

Proposition 5.1 (Smooth-field asymptotics). *For a periodic shape-regular triangulation family,*

$$\mathcal{B}_a[S_a n] = \frac{R_a}{8} \int_{\Omega} Q_{\mathcal{T}}(\nabla n) dx + O\left(R_a \left(a + \frac{a^2}{d_a}\right)\right), \quad (5.1)$$

where $Q_{\mathcal{T}}$ is the sum of $|D_v n|^4$ with the asymptotic edge direction densities of the selected mesh.

For the AP-E8 uniform positive Freudenthal split,

$$Q_{6,+}(A) = \sum_{i=1}^3 |Ae_i|^4 + \sum_{1 \leq i < j \leq 3} |A(e_i + e_j)|^4 + |A(e_1 + e_2 + e_3)|^4. \quad (5.2)$$

For the checkerboard five-tetrahedron mesh, each face-diagonal sign occurs at density one half and there is no body diagonal, so

$$Q_5(A) = \sum_{i=1}^3 |Ae_i|^4 + \frac{1}{2} \sum_{1 \leq i < j \leq 3} (|A(e_i + e_j)|^4 + |A(e_i - e_j)|^4). \quad (5.3)$$

Now set $\gamma_a = ca^{-p}$ and first keep $d_a = d > 0$ fixed. Then

$$w_a = ca^{1-p}, \quad R_a = \frac{c}{d^2} a^{2-p}. \quad (5.4)$$

Corollary 5.2 (Unique edge-stable, smooth-vanishing window). *The bounded sublevels have a uniform relative edge margin precisely for $p \geq 1$, while the barrier vanishes on every smooth field precisely for $p < 2$. Thus the only power window with both properties is*

$$\boxed{1 \leq p < 2.} \quad (5.5)$$

The minimal choice is $p = 1$, namely $\gamma(a) = \bar{\gamma}/a$, for which $w_a = \bar{\gamma}$ and a smooth barrier is $O(a)$.

More generally, if $d_a \sim d_0 a^\alpha$, the two requirements become

$$p \geq 1, \quad p + 2\alpha < 2. \quad (5.6)$$

The window is nonempty only for $\alpha < 1/2$. An absolute edge margin also requires $\alpha = 0$. We therefore retain the fixed value $\epsilon_a = 0.01$ and do not send ϵ_a either to 1 or to the no-zero threshold $-1/3$.

6 Barrier regimes, the zero Gamma limit, and the no-free-lunch theorem

Let $P_a n$ be the dual-cell piecewise-constant extension of the nodal field and $I_a n$ its tetrahedral affine interpolant. On the explicitly declared state space $L^2(\Omega; \mathbb{R}^4)$ define

$$\mathfrak{B}_a(u) = \begin{cases} \mathcal{B}_a[n], & u = P_a n \text{ for an admissible nodal field,} \\ +\infty, & \text{otherwise.} \end{cases} \quad (6.1)$$

Boundary-node values have zero volume measure, so a zero-energy strong- L^2 limit cannot retain a trace unless a coercive base term is also present.

Theorem 6.1 (Rigorous barrier regimes). *Fix $d = 1 - \epsilon > 0$ and a periodic shape-regular triangulation family.*

(i) *If $R_a \rightarrow 0$, then in strong L^2*

$$\mathfrak{B}_a \xrightarrow{\Gamma} \mathfrak{B}_0(u) := \begin{cases} 0, & u \in L^2(\Omega; \mathbb{S}^3), \\ +\infty, & \text{otherwise.} \end{cases} \quad (6.2)$$

This limit forgets both trace and degree.

(ii) *If $\inf_a R_a > 0$, every bounded barrier sublevel gives a uniform discrete $W^{1,4}$ bound on $I_a n$. The affine fields are precompact in $C^{0,\beta}$ for every $\beta < 1/4$, and normalized-affine degree is closed.*

- (iii) If $R_a \rightarrow \infty$, the $W^{1,4}$ seminorm tends to zero on a bounded sublevel. With the fixed vacuum boundary trace, the only possible limit is e_0 .
- (iv) If $R_a \rightarrow R \in (0, \infty)$, Eq. (5.1) identifies the Cauchy–Born density on fixed smooth sampled fields. It does not by itself identify the Gamma limit. For a periodic multi-cell stencil the candidate integral density must be obtained from the corresponding homogenized cell problem; oscillatory correctors may lower the raw $Q_{\mathcal{T}}$. That cell formula is not evaluated here.

Proof. For (i), strong L^2 limits of the sphere-valued piecewise-constant fields remain sphere-valued almost everywhere, giving the $+\infty$ part and the nonnegative liminf. Smooth sphere-valued maps which equal the vacuum in a shrinking boundary collar are dense in $L^2(\Omega; \mathbb{S}^3)$. For each fixed smooth map, nodal sampling converges strongly and Eq. (5.1) makes the barrier vanish. A slow diagonal choice gives the recovery sequence.

For (ii), Eq. (2.5) on the tetrahedral edges and shape regularity give

$$\|\nabla I_a n\|_{L^4}^4 \leq \frac{C}{R_a} \mathcal{B}_a[n]. \quad (6.3)$$

Moreover $\|P_a n - I_a n\|_{L^4} \leq C a \|\nabla I_a n\|_{L^4}$, so both extensions have the same limit. Morrey compactness gives the asserted uniform convergence, which preserves the no-zero interpolation and degree. If $R_a \rightarrow \infty$, Eq. (6.3) makes the seminorm vanish; the fixed trace and Poincaré’s inequality select the vacuum. Statement (iv) is precisely the boundary of Eq. (5.1): periodic discrete-to-continuum identification requires minimization over cell correctors, not merely evaluation on the uncorrected affine sample. $\square$

Theorem 6.2 (Fixed- ϵ no-free-lunch theorem). *No scaling of the AP-E8 logarithmic barrier with fixed $\epsilon \in (-1/3, 1)$ makes it both*

- (a) *vanish on every smooth continuum field, and*
- (b) *by itself provide compactness strong enough to close continuum degree sectors.*

At the critical compactness scaling $p = 2$, the raw Cauchy–Born consistency tensors for the six- and five-tetrahedron meshes cannot be made equal by multiplying one of them by a constant. Equality or inequivalence of their homogenized Gamma densities is a separate, unevaluated cell problem.

Proof. Smooth vanishing on every smooth field is equivalent to $R_a \rightarrow 0$, whereas the $W^{1,4}$ estimate from the barrier is uniform only when $\liminf R_a > 0$. Failure of degree closure in the vanishing regime is explicit. Choose a smooth map $U : B_1 \rightarrow \mathbb{S}^3$ of degree one which equals e_0 on a collar of ∂B_1 , and put

$$n^{(r)}(x) = \begin{cases} U((x - x_0)/r), & x \in B_r(x_0), \\ e_0, & x \notin B_r(x_0). \end{cases}$$

It has degree one and

$$\int |\nabla n^{(r)}|^2 = O(r), \quad \int |\nabla n^{(r)}|^4 = O(r^{-1}).$$

For a resolved nodal sample, a/r sufficiently small makes every edge admissible. Equation (2.4) and edge counting in B_r then give $\mathcal{B}_a[S_a n^{(r)}] \leq C R_a / r$. Choose $r_a = \max\{R_a^{1/2}, a^{1/2}\}$. Then $r_a \rightarrow 0$, $r_a/a \rightarrow \infty$, and $R_a/r_a \rightarrow 0$. The sampled normalized-affine maps therefore have degree one and bounded, indeed vanishing, barrier energy, but converge strongly in L^2 to the degree-zero vacuum. Thus a vanishing barrier cannot close degree sectors.

For the final assertion, take a linear gradient with $Ae_1 = v$, $Ae_2 = Ae_3 = 0$. Equations (5.2)–(5.3) give $Q_{6,+} = 4|v|^4$ and $Q_5 = 3|v|^4$. If a scalar rescaling matched these, it would use a factor $4/3$. Instead take $Ae_1 = Ae_2 = v$, $Ae_3 = 0$; then $Q_{6,+} = 36|v|^4$ and $Q_5 = 12|v|^4$, whose ratio is 3. Hence the two uncorrected smooth-sampling quartics are not proportional. $\square$

Remark 6.3 (What the no-go does not prove). The degree-one bubble has base Skyrme energy of order $1/r$. Consequently the preceding construction concerns the barrier alone and does not prove that bounded sublevels of the *complete* Skyrme action lose degree. Whether the base Skyrme minors themselves supply topological-current compactness is exactly the remaining theorem-level blocker.

7 Conditional Gamma theorem for the complete action

Theorem 7.1 (Stable addition of a vanishing barrier). *Assume that the AP-E7 base functionals E_a satisfy, in strong $L^2 \times L^2$,*

$$(i) \ E_a \xrightarrow{\Gamma} E_0;$$

(ii) *their sublevels are equicoercive;*

(iii) *for every $u = (n, s)$ with $E_0(u) < \infty$, there are smooth pairs $u^k = (n^k, s^k)$, with the declared boundary data, such that*

$$u^k \rightarrow u \text{ in } L^2 \times L^2, \quad E_0(u^k) \rightarrow E_0(u); \quad (7.1)$$

(iv) *for every fixed k , the nodal samples $S_a u^k$ recover the base energy:*

$$S_a u^k \rightarrow u^k, \quad E_a(S_a u^k) \rightarrow E_0(u^k). \quad (7.2)$$

If

$$\frac{a^2}{d_a} \longrightarrow 0, \quad R_a = \frac{\gamma_a a^2}{d_a^2} \longrightarrow 0, \quad (7.3)$$

then

$$E_a + \mathcal{B}_a \xrightarrow{\Gamma} E_0. \quad (7.4)$$

The full family inherits $L^2 \times L^2$ equicoercivity from E_a .

Proof. The liminf follows from $\mathcal{B}_a \geq 0$. For every fixed k , Eq. (5.1) and Eq. (7.3) give $\mathcal{B}_a[S_a n^k] \rightarrow 0$; the constants hidden in the expansion may depend on k but not on a . First take $a \rightarrow 0$ at fixed k , then choose a slow diagonal $k = k(a) \rightarrow \infty$. Equations (7.1) and (7.2) give the required full recovery sequence. Finally, $E_a \leq E_a + \mathcal{B}_a$, so every full-action sublevel lies in the corresponding equicoercive base sublevel. $\square$

This theorem is unconditional as an abstract stability statement, but its four hypotheses have not yet been proved for the specific one-corner forward Skyrme discretization. The missing statements are:

1. a discrete compensated-compactness and liminf theorem for the three forward Skyrme minors;
2. smooth fixed-boundary, fixed-degree density in the natural class

$$\{n \in W^{1,2} : \partial_i n \wedge \partial_j n \in L^2\};$$

3. exclusion or accounting of a topological-current concentration defect;
4. translation tightness when $L \rightarrow \infty$.

Without centering or quotienting translations, joint $a \downarrow 0$, $L \uparrow \infty$ equicoercivity is false: a localized soliton can translate to infinity while retaining its energy and norm.

8 Alternative triangulations and translated starts

The numerical card uses exactly the same tetrahedral complex for its barrier edges and its normalized-affine degree. It tests:

1. the original uniform six-tetrahedron Freudenthal mesh;
2. both phases of the conforming checkerboard five-tetrahedron mesh, whose central parity is phase $+i + j + k \pmod{2}$;
3. starts translated by noninteger lattice offsets before imposing the exact vacuum boundary;
4. wider fixed- L spacing and fixed- a volume sequences;
5. the matched trajectories $p = 0, 1, 2$, normalized so that $\gamma(0.4) = 1$.

The five-tetrahedron mesh has one central tetrahedron of determinant magnitude two and four corner tetrahedra of determinant magnitude one in every cube. The checkerboard parity flip makes every shared face diagonal agree. Exact combinatorial tests check nonzero volume, spatial orientation, face multiplicity, Euler characteristic one, and degree on regular targets.

For each relaxed background, the card records total, base, and barrier energies; direct tangent-plus-scalar stationarity; minimum edge dot; degree at three generic regular targets; the $1 - n_0$ barycentre and centered radius; an absolute-baryon-density proxy radius; and high-quantile local gradient concentration. A translated start is therefore not silently recentered in the reported observables.

9 Numerical verdict

The production values in this section are generated by

`route_f/code/scan_ap_e9_gamma_triangulation.py.`

The machine-readable and human-readable cards are

`route_f/output/ap_e9_gamma_triangulation.json,`
`route_f/output/ap_e9_gamma_triangulation.md.`

All 23 independently relaxed cases are stationary within the declared 2×10^{-6} projected-gradient-density tolerance, remain admissible, and return degree $(1, 1, 1)$ at the three regular targets. The largest gradient density is 1.7224×10^{-6} and the smallest pair margin is 0.067786. The smooth compact-support test gives

p	fitted barrier slope	predicted $2 - p$	R^2
0	2.159102	2	0.999141
1	1.159102	1	0.997025
2	0.159102	0	0.863298

Thus the fixed-profile calculation detects the predicted smooth-sampling exponent split. It does not imply that a relaxed branch has entered that asymptotic regime. For the matched $p = 1$, fixed- $L = 6$ branch the production sequence is

a	E_a	$\mathcal{B}_a$	$\mathcal{F}_a$	centered RMS	gradient density
0.428571	55.307957	18.487241	73.795198	1.298396	2.36×10^{-7}
0.375000	55.677145	17.909776	73.586921	1.261284	1.94×10^{-7}
0.333333	56.448068	17.495634	73.943702	1.232911	1.08×10^{-6}
0.300000	57.507365	17.148880	74.656246	1.211449	2.28×10^{-7}
0.272727	58.801673	16.793372	75.595045	1.196040	3.21×10^{-7}
0.250000	60.320889	16.351169	76.672058	1.186597	5.16×10^{-7}

The finest-three total-energy and radius spreads are 2.629% and 2.051%, respectively, but the fitted barrier slope is only 0.2196, outside the predeclared 1.0 ± 0.4 window. Hence the relaxed fields have not demonstrated the predicted $O(a)$ barrier disappearance.

At fixed $a = 0.4$, increasing the box gives

L	N	$\mathcal{F}_a$	centered RMS	pair margin
6	16	73.627393	1.274459	0.080722
8	21	73.346860	1.425707	0.085418
10	26	73.289479	1.510286	0.085975
12	31	73.273437	1.558499	0.086231

The largest-three energy spread is only 0.1001%, whereas the radius spread is 8.521%, exceeding its 5% gate. Energy stabilization alone therefore does not establish volume stabilization.

The alternative-mesh screen was repeated at two spacings, with both centered and translated starts:

a	mesh	centered $\mathcal{F}_a$	translated $\mathcal{F}_a$	centered RMS	pair margin
0.4	six-tet	73.627393	73.614920	1.274459	0.080722
0.4	five-tet phase 0	69.630225	69.630225	1.210207	0.085973
0.4	five-tet phase 1	69.620878	69.620878	1.212765	0.086169
0.3	six-tet	74.656246	74.656246	1.211449	0.109410
0.3	five-tet phase 0	70.659350	70.659350	1.128126	0.100946
0.3	five-tet phase 1	70.658169	70.658169	1.128277	0.101019

Translation changes the terminal energy negligibly, but six-versus-five-tet energy spreads remain 5.442% at $a = 0.4$ and 5.355% at $a = 0.3$, above the 3% gate. They are not visibly shrinking. On the common $N = 21, L = 6$ grid, the two smooth-vanishing controls $p = 0$ and $p = 1$ give $\mathcal{F}_a = 69.844123$ and 74.656246: their 6.446% spread also exceeds the 3% regulator gate, although their radius spread 3.897% passes.

Consequently the strict card passes all eight implementation and provenance checks but returns

`numerical_regulator_stability=false,      continuum_background=false.`

The failure is informative rather than an optimizer failure: every required stationary, admissibility, and degree check passes, while four independent continuum diagnostics fail. Irrespective of that empirical verdict, a finite collection of convergent observables could not replace the missing Skyrme-minor liminf and topological-current compactness proof.

10 Recent literature and exact transfer boundary

No located paper proves the three-dimensional $\mathbb{S}^3$ -valued AP-E8 functional with its logarithmic edge barrier. The closest recent rigorous work proves charge-measure compactness and Gamma convergence for a two-dimensional $\mathbb{S}^2$ baby-Skyrme setting [1]; it is a proof architecture, not a theorem that transfers dimensions and targets. Recent sphere-valued finite-element and triangulation analyses identify the need for shape regularity, consistent weights, and stability hypotheses [2, 3, 4]. Scaling thresholds can change Gamma limits in other spin systems [5], but that result does not contain this barrier.

The checkerboard five-tetrahedron construction and the warning that a lattice cutoff can be part of the model have a direct three-dimensional predecessor [6]. Earlier geometric degree and admissibility ideas [7, 8] motivate the design but do not imply a continuum theorem here. Three-dimensional lattice Skyrmin branches are known to be numerically sensitive near continuum regimes [9], which is why translated starts and branch diagnostics are mandatory rather than cosmetic.

11 Fail-closed decision and ordered continuation

Fail-closed gate 11.1 (Hessian authorization). The full same-action Riemann Hessian and regulated determinant variation may start only after all of the following are true:

1. one $p < 2$ background sequence is stationary and stable under spacing, volume, triangulation, translation, and regulator changes;
2. the barrier fraction vanishes with the predicted scaling;
3. the base Skyrme-minor liminf and fixed-degree recovery problem is closed, or an explicitly modified continuum action is adopted;
4. translation tightness is fixed by a declared quotient or a physical centering condition, not by an optimization anchor.

Until then:

```
regulator_stable_continuum_background = false,  
same_action_hessian_authorized = false.
```

and the degree-one Route-E portal remains prohibited. The finite GW/domain-wall kernel and the actual $SO(3)$ mapping-torus mod-two index remain parallel lanes. The mathematically preferred next choices are:

1. prove compactness of the discrete Skyrme topological current in the natural minor-bounded class;
2. evaluate the periodic finite- R homogenized cell problem before identifying any raw $Q_{\mathcal{T}}$ with a Gamma density;
3. if the base-minor route fails, adopt the critical $p = 2$ term as a declared physical quartic and replace the raw stencil by a rotationally calibrated, triangulation-independent discretization;
4. continue the $p = 1$, fixed- ϵ numerical sequence only as a vanishing-regulator candidate, never as a proof of degree closure.

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